Mathmatical Analysis

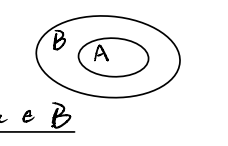
- Completeness of $\mathbb{R}$
- Uniform continuity
- Uniform convergence

Chapter 1 : Real Number ($\mathbb{R}$)

§ 1.1 Concepts:

① $\mathbb{R} = (-\infty, \infty)$
$\emptyset$ = empty set
A is a set, a is an element : $a \in A$ or $a \notin A$

② $A \subseteq B$: A is a subset of B



def: for any $a \in A$, we have $a \in B$

③ $\forall$: for any
$\exists$: there exist

④ $A \subset B$ if $A \subseteq B$ and "A is a proper subset of B"
⇒ "A is a proper subset of B" ⇒ there is $b \in B$ where $b \notin A$

⑤ $A = B$: $A \subseteq B$ and $B \subseteq A$

Set operation

":=" ⇔ definition

A, B :
$A \cup B := \{x : x \in A \text{ or } x \in B\}$
$A \cap B := \{x : x \in A \text{ and } x \in B\}$
$A \setminus B := \{x : x \in A \text{ and } x \notin B\}$

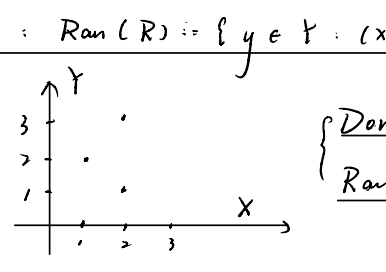
Cartisian Product:

X, Y are sets. $X \times Y := \{(x,y) : x \in X, y \in Y\}$
$(x,y), (z,w) \in X \times Y$, $(x,y) = (z,w) \Leftrightarrow x = z$ and $y = w$.
※ $\emptyset \times Y = \emptyset$

Function: let X and Y be sets. Any subset $R \subseteq X \times Y$ is called relation on $X \times Y$

domain : $\mathrm{Dom}(R) := \{x \in X : (x,y) \in R \text{ for some } y \in Y\} \subseteq X$
range : $\mathrm{Ran}(R) := \{y \in Y : (x,y) \in R \text{ for some } x \in X\} \subseteq Y$

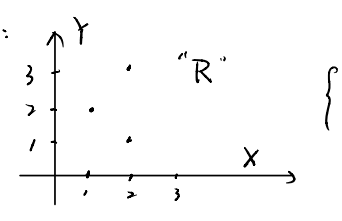
e.x. :



$\mathrm{Dom}(R) = \{1, 2\}$
$\mathrm{Ran}(R) = \{1, 2, 3\}$

☆ Def : a function f from X into Y is a relation on $X \times Y$ such that $\mathrm{Dom}(R) = X$ and $\forall x \in X$, there is a unique $y \in Y$ with $(x,y) \in f$

example :



$\mathrm{Dom}(R) = \{1, 2\}$
$\mathrm{Ran}(R) = \{1, 2, 3\}$

Is relation R a fuction ?
No! since :
1. $\mathrm{Dom}(R) \neq X$.
2. $\exists a \in X$, there exist more than one $y \in Y$, $(x,y) \in R$.

Notation

Suppose f is a function from X into Y
We write $f : X \to Y$ or $x \mapsto y$
if $(x,y) \in f$, we call y the value of f at x. notion: $y = f(x)$
or $f : x \mapsto y$, we call x is a preimage of y
if $f : X \to Y$, we say f is defined on X and Y is the codomain of f
$\mathrm{Ran}(f) = \{y \in Y : y = f(x) \text{ for some } x \in X\}$
$= \{f(x) : x \in X\} \subseteq Y$.
$y \in \mathrm{Ran}(f) \Leftrightarrow \exists x \in X, y = f(x)$

Inverse function:
if $\mathrm{Ran}(f) = Y$ and for any $y \in Y$ there is a unique $x \in X$ such that $f(x) = y$.
then we say $f : X \to Y$ has an inverse
and define $f^{-1} : Y \to X$, $y \mapsto x$

* two function f, g are the same if $\mathrm{Dom}(f) = \mathrm{Dom}(g)$ ① and $\mathrm{Ran}(f) = \mathrm{Ran}(g)$ ② and $\forall x, f(x) = g(x)$ ③

Real fuctions : $f : \mathbb{R} \to \mathbb{R}$

i) $e^x : \mathbb{R} \to (0, \infty)$, $\log x : (0, \infty) \to \mathbb{R}$ } inverse.
ii) trig function : sin x, cos x, tan x, cot x ...
iii) power fuction : x^α, $x > 0$, $\alpha \in R$

1.2 Ordered-field axioms

What is $\mathbb{R}$?

define : $\mathbb{N} = \{1, 2, 3 \dots\}$. Natural Numbers
$\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$ Integers
$\mathbb{Q} = \{\frac{m}{n} : m, n \in \mathbb{Z}, n \neq 0\}$. rational numbers.

$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$ "$\sqrt{2} \notin \mathbb{Q}$"
$\mathbb{Q}^c = \mathbb{R} / \mathbb{Q}$ "Set of irrational numbers"

What is $\mathbb{R}$?

① $\mathbb{R}$ is a field satisfying field axiom:
Field axiom : '+, −, ×' : these operations satisfy natural properties
$a + b = b + a$, $(a+b)+c = a+(b+c)$, $a \neq 0$: $a \cdot a^{-1} = 1$

② There is an order "<" on $\mathbb{R}$ that is the order axiom:
$\forall a, b \in \mathbb{R}$. $a = b$ or $a > b$ or $a < b$
natural properties : $a > b$:
- $a + c > b + c$ $\forall c \in \mathbb{R}$
- $ac > bc$ $\forall c \in \mathbb{R}$ $c > 0$.
- $ac < bc$ $\forall c \in \mathbb{R}$ $c < 0$

$a > b$, $b > c \Rightarrow a > c$
$a \leq b \Leftrightarrow a < b$ or $a = b$.

remark : $\mathbb{N}, \mathbb{Z}$ Satisfies :
① $m, n \in \mathbb{Z} \Rightarrow m + n, mn \in \mathbb{Z}$
② $n \in \mathbb{Z}$ then $n \in \mathbb{N} \Leftrightarrow n \geq 1$
③ there is no $n \in \mathbb{Z}$ such that $0 < n < 1$ (i.e. $\mathbb{Z} \cap (0,1) = \emptyset$)

Absolute Value

$a \in \mathbb{R}$ $|a| := \begin{cases} a, & a \geq 0. \\ -a, & a < 0 \end{cases}$

Property :
(1) $|a \cdot b| = |a| \cdot |b|$
(2) $|a| \leq M \Leftrightarrow -M \leq a \leq M$
(3) $|a| = |-a| \Rightarrow |a - b| = |b - a|$
(4) ① $|a + b| \leq |a| + |b|$ "triangle inequality"
② $||a| - |b|| \leq |a - b|$.

proof : $|a| = |b + a - b| \leq |b| + |a - b|$
thus: $|a| - |b| \leq |a - b|$ ①
interchanging a, b : $|b| - |a| \leq |b - a| = |a - b|$
⇒ $|a| - |b| \geq -|a - b|$ ②
By ①, ② : $-|a - b| \leq |a| - |b| \leq |a - b|$
therefore $||a| - |b|| \leq |a - b|$

Interval on $\mathbb{R}$

$a, b \in \mathbb{R}$
$[a, b] = \{x \in \mathbb{R} : a \leq x \leq b\}$.
$(a, b) = \{x \in \mathbb{R} : a < x < b\}$.
$(-\infty, b) = \{x \in \mathbb{R} : x < b\}$.
$(a, \infty) = \{x \in \mathbb{R} : x > a\}$.

§ 1.3 Completeness Axiom

Def : $\emptyset \neq E \subseteq \mathbb{R}$. (E is a nonempty subset of $\mathbb{R}$)
i) E is called bounded above ⇔ $\exists M \in \mathbb{R}$ such that $\forall a \in E \Rightarrow a \leq M$.
e.g. $E = (1, 5)$. then $M = 7$ is an upper bound, E is bounded above.

Remark : ① if M is an UB. so is M+1 ⇒ Upper Bound is not unique if exists.
② UB may not exist (e.g. $E = \mathbb{R}$)

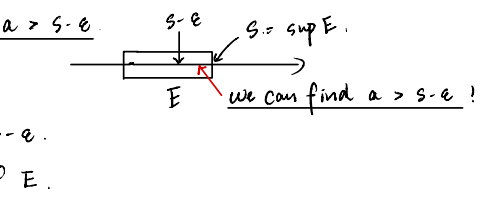
ii) A number $s \in \mathbb{R}$ is called a supremum of E
if and only if : ① s is an upper bound of E
② $s \leq M$ for any upper bound M of E (the smallest UB)
if E has a supremum s. We write $s = \sup E$

e.x. $E = [0, 1]$ prove that $\sup E = 1$
proof : $\forall x \in [0, 1]$, $x \leq 1$ so 1 is an UB.
let M be an UB of E. then $M \geq x$ $\forall x \in [0, 1] \Rightarrow M \geq 1$
∴ $\sup E = 1$ (prove by definition)

Remark : ① if E has a supremum. then it has only one supremum (uniqueness)
proof : if E has two supremum s, t.
then by def of supremum : $s \leq t$ and $t \leq s$, therefore : $s = t$

"Approximation Property"

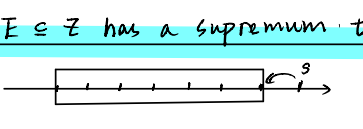
Theorem : if E has a supremum. let $s = \sup E$ ($s - \varepsilon < a \leq \varepsilon$)
Then $\forall \varepsilon > 0$ $\exists a \in E$ such that $a > s - \varepsilon$



Proof : suppose not. then :
"prove by negation" ☆
$\exists \varepsilon > 0. \forall a \in E, a < s - \varepsilon$.
then $s - \varepsilon$ is an UB of E.
Since $s = \sup E$. and $s - \varepsilon < s$. Contradiction !

Theorem : if $E \subseteq \mathbb{Z}$ has a supremum. then $\sup E \in E \subseteq \mathbb{Z}$

proof :



"if sup E is not an integer. we can always shift sup E left"

Completeness Axiom : $\emptyset \neq E \subseteq \mathbb{R}$. Must claim : Unempty
"if E is bounded above. Then E has a supremum i.e. $\sup E \in \mathbb{R}$"

Archimedian Principle $\forall a, b \in \mathbb{R}$ $a > 0$. then $\exists n \in \mathbb{N}$ $b < na$.
(i.e. na can be arbitrarily large as $n \in \mathbb{N}$)

Proof : if $b < a$. set $n = 1$.
if $b \geq a$. Consider $E = \{k \in \mathbb{N} : ka \leq b\}$. : ① $1 \in E \neq \emptyset$ since $a \leq b$.
② $\forall k \in E \Rightarrow ka \leq b \Rightarrow k \leq \frac{b}{a} \in \mathbb{R}$.
⇒ $\frac{b}{a}$ is an upper bound of E
By completeness theorem. $\exists s \in \mathbb{Z}$, $s = \sup E$
set $n = s + 1 \notin E$. then $na > b$, proved.

$k \leq \frac{b}{a} \Rightarrow UB \in \mathbb{R}$.
⇒ ∃ sup E . let $n = \sup E + 1$

② ①

"Completeness axiom ⇒ density rationals."

e.x. $B = \{1 - \frac{1}{n}, n \in \mathbb{N}\}$ prove that $\sup B = 1$.
proof : (1) since $\forall a \in B$, $a \leq 1$. 1 is an upper bound of B.
(2). let M be UB of B., we need to show that $M \geq 1$.
if $M < 1$: By archimedian principle : $\exists n \in \mathbb{N}$: $n > \frac{1}{1-M}$ ($a = 1$, $b = \frac{1}{1-M} \Rightarrow na > b \Rightarrow n > \frac{1}{1-M}$)
$n > \frac{1}{1-M} \Rightarrow 1 - \frac{1}{n} > M$. M is not upper bound of B

Lower Bound and infima

① The set E is bounded below if there exists $m \in \mathbb{R}$ such that : $m \leq a$ for every $a \in E$
such m is a lower bound of E.
② A number $t \in \mathbb{R}$ is an infimum of E if:
1) t is a lower bound of E
2) t is the largest lower bound : $m \leq t$ for every lower bound m of E
written $t = \inf E$ if it exists

Suprema, infima, negation
① def : $-E := \{-x : x \in E\}$
② E has a supremum ⇔ −E has a infimum.
③ E has a infimum ⇔ −E has a supremum

Proof : if E has a supremum say $s = \sup E$
We need to show that $\inf(-E) = -s$
1. $\forall x \in -E$, $-x \in E$, $-x \leq s \Rightarrow x \geq -s$., $-s$ is a lower bound
2. let m be a lower bound of −E ⇒ −m is a upper bound of E.
$-m \geq \sup E = s. \Rightarrow m \leq -s$
Hence −s is a infimum of −E

Theorem 1.21 : monotonicity under inclusion

Let $A \subseteq B$ be nonempty subset of $\mathbb{R}$:
(i) if B has a supremum then A has a supremum and : $\sup A \leq \sup B$.
(ii) if B has a infimum then A has a infimum and : $\inf A \geq \inf B$.
proof : (i) Every upper bound of B is a upper bound of A ⇒ sup B is an UB of A ⇒ $\sup A \leq \sup B$
(ii) $A \subseteq B \Rightarrow -A \subseteq -B. \Rightarrow \sup(-A) \leq \sup(-B) \Rightarrow \inf(A) \geq \inf(B)$

The extended real number
Define : $\overline{\mathbb{R}} := \mathbb{R} \cup \{-\infty, +\infty\}$
For $\emptyset \neq E \subseteq R$: $\sup E = +\infty$ if E is unbounded above
$\inf E = -\infty$ if E is unbounded below
For $E = \emptyset$: $\sup E = -\infty$. $\inf E = \infty$.
⇒ For $A \subseteq B \subseteq \overline{\mathbb{R}}$: $\sup A \leq \sup B$. $\inf A \geq \inf B$.

if $E \neq \emptyset$. E is at most countable ⇔ there is a surjection $\mathbb{N} \to E$
proof : ① if E is at most countable :
if E is countable there is bijection $\mathbb{N} \to E$, hence surjection $\mathbb{N} \to E$.
if E is finite : say $E = \{e_1, e_2, \dots, e_m\}$. , map $1 \dots m$ to each element
and send every later integer to e_m
⇒ surjection : $\mathbb{N} \to E$.
② if there is surjection $\mathbb{N} \to E$:
retain the first index at each value of E occurs.
⇒ a subset of $\mathbb{N}$ ⇒ form a bijection onto E.
- if the subset is finite. E is finite
- if the subset is countable. E is countable

Mathmatical induction

Theorem 1.22. well-ordering principle.
Every subset $E \subseteq \mathbb{N}$ has a least element : $\inf E \in E$
proof : Since $E \subseteq \mathbb{N}$. $-E := \{-n : n \in E\}$ is a nonempty subset of $\mathbb{Z}$
By completeness axiom : Since $-E \subseteq \mathbb{R}$. −E has a supremum $\in \mathbb{R}$.
By Theorem proved previously (if $E \subseteq \mathbb{Z}$ has a supremum then $\sup E \in \mathbb{Z}$).
∴ $\sup(-E) \in -E. \Rightarrow -\sup(-E) \in E \Rightarrow \inf E \in E$.

Theorem : principle of mathmatical induction
let A(n) be a proposition ($n \in \mathbb{N}$)
if : ① A(1) is true and.
② for $n \in \mathbb{N}$ if A(n) is true ⇒ A(m) is true $\forall n \in \mathbb{N}$.
proof : suppose the conclusion is false :
$E := \{n \in \mathbb{N} : A(n) \text{ is false}\}$ $E \neq \emptyset$.
$m := \min E$ exists by well-ordering theorem
A(1) is true. ⇒ $m \neq 1$ ⇒ $m - 1 \in \mathbb{N}$ ⇒ A(m−1) is true
Since $m = \min E$. $m - 1 \notin E$.
A(m−1) is true ⇒ A(m) is true. which is a contradiction

Summation notation
$\sum_{k=j}^{n} x_k = x_j + x_{j+1} + \cdots + x_n$

Binomial Coefficients
$n! := n(n-1) \cdots 2 \cdot 1$
Binomial coefficients
$\binom{n}{k} := \frac{n!}{(n-k)!\,k!}$ in particular $\binom{0}{0} = 1$ $\binom{n}{0} = \binom{n}{n} = 1$

Pascal's identity
$\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}$
$\frac{n!}{(n-k+1)(k-1)!} + \frac{n!}{(n-k)!\,k!} = \frac{n! \cdot k}{(n+1-k)\,k!} + \frac{n! \cdot (n-k+1)}{(n-k+1)!\,k!}$
$= \frac{n!(n+1)}{(n+1-k)\,k!} = \binom{n+1}{k}$

Binomial Theorem:
$(a+b)^n = \sum_{n=0}^{n} \binom{n}{k} a^k b^{n-k}$

Inverse functions
(1) one-to-one : $\forall a, b \in X$ $a \neq b \Rightarrow f(a) \neq f(b)$
(2) onto : $\forall b \in Y$. $\exists a$. $f(a) = b$.
(3) bijective : one-to-one & onto

Thm1.30 : The following are equivalent:
① $f : X \to Y$ has an inverse function $f^{-1}: Y \to X$.
② f is a bijection
③ There exists a function $g : Y \to X$ such that : $g \circ f = id_X$, $f \circ g = id_Y$
"$x = g(f(x))$ $y = f(g(y))$"

①→③ : if f^{-1} exist. take $g = f^{-1}$. then $f \circ f^{-1} = id_X$. ($f(f^{-1}(x)) = x$)
$f^{-1} \circ f = id_Y$ ($f^{-1}(f(y)) = y$)
③→② : if $f(x_1) = f(x_2)$ then $g(f(x_1)) = x_1$, $g(f(x_2)) = x_2$
⇒ $x_1 = x_2$ f is injective
Also. $\forall y \in Y$. $y = f(g(y))$ ∴ f is surjective.
⇒ f is bijective
②→① : if f is a bijection. $\forall y \in Y$. y has a unique preimage $x \in X$.
then we can define $f^{-1}(y) = x$

Image and Preimage
image of $E \subseteq X$: $f(E) = \{f(x) : x \in E\}$.
preimage of $V \subseteq Y$ $f^{-1}(V) = \{x : f(x) \in V\}$

Indexed families of sets
$\{E_1, E_2, E_3, E_4\} = \{E_\alpha\}_{\alpha \in A}$
$\bigcup_{\alpha \in A} E_\alpha = E_1 \cup E_2 \cup E_3 \cup E_4$.
$\bigcap_{\alpha \in A} E_\alpha = E_1 \cap E_2 \cap E_3 \cap E_4$

De Morgan's law
$(\bigcup_{\alpha \in A} E_\alpha)^c = \bigcap_{\alpha \in A} E_\alpha^c$ $(\bigcap_{\alpha \in A} E_\alpha)^c = \bigcup_{\alpha \in A} E_\alpha^c$

Theorem : images and preimages of set operations
$f : X \to Y$: (i) For $E_\alpha \subseteq X$, $f : X \to Y$
$f(\bigcup_{\alpha \in A} E_\alpha) = \bigcup_{\alpha \in A} f(E_\alpha)$, $f(\bigcap_{\alpha \in A} E_\alpha) \subseteq \bigcap_{\alpha \in A} f(E_\alpha)$ not "="
(ii) For $B, C \subseteq X$.
$f(C) / f(B) \subseteq f(C / B)$ ☆ not =.
(iii) For subsets $V_\alpha \subseteq Y$:
$f^{-1}(\bigcup_{\alpha \in A} E_\alpha) = \bigcup_{\alpha \in A} f^{-1}(E_\alpha)$ $f^{-1}(\bigcap_{\alpha \in A} E_\alpha) = \bigcap_{\alpha \in A} f^{-1}(E_\alpha)$
proof : $\forall y \in f(C) \setminus f(B)$ choose $c \in C$ and $f(c) = y$. , we have $c \notin B$.
⇒ $c \in C \setminus B$, $y \in f(C \setminus B)$

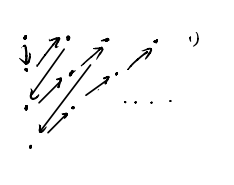
Countable & Uncountable
Let E be a set :
(i) E is finite if $E \neq \emptyset$, and for some $n \in \mathbb{N}$ there is a bijection $\{1, 2, \dots, n\} \to E$.
(ii) E is countable if there is a bijection $\mathbb{N} \to E$.
(iii) E is at most countable if finite or countable
(iv) E is uncountable if neither finite nor countable

Cantor's Theorem
(0,1) is uncountable
proof : Suppose (0,1) is countable. then $\exists f : \mathbb{N} \to (0,1)$ is bijection.
$(0,1) = \{f(1), f(2), \dots\}$.
Write each f(n) in decimal form :
$f(n) = 0.d_{n1} d_{n2} d_{n3} \cdots$, $n_j \in \{0, 1, \dots, 9\}$.
$f(1) : 0.\boxed{d_{11}} d_{12} d_{13} d_{14} \cdots$
$f(2) : 0.d_{21} \boxed{d_{22}} d_{23} d_{24} \cdots$
$f(3) : 0.d_{31} d_{32} \boxed{d_{33}} d_{34} \cdots$
$f(4) : 0.d_{41} d_{42} d_{43} \boxed{d_{44}} \cdots$
⋮
let $\beta_k = \begin{cases} d_{kk} + 1 & d_{kk} \leq 5 \\ d_{kk} - 1 & d_{kk} > 5 \end{cases}$
let $x = 0.\beta_1 \beta_2 \beta_3 \cdots$
$x \in (0,1)$ and x differs from every f(n)
⇒ $x \notin \{f(n) : n \in \mathbb{N}\}$ Contradiction to Surjectivity

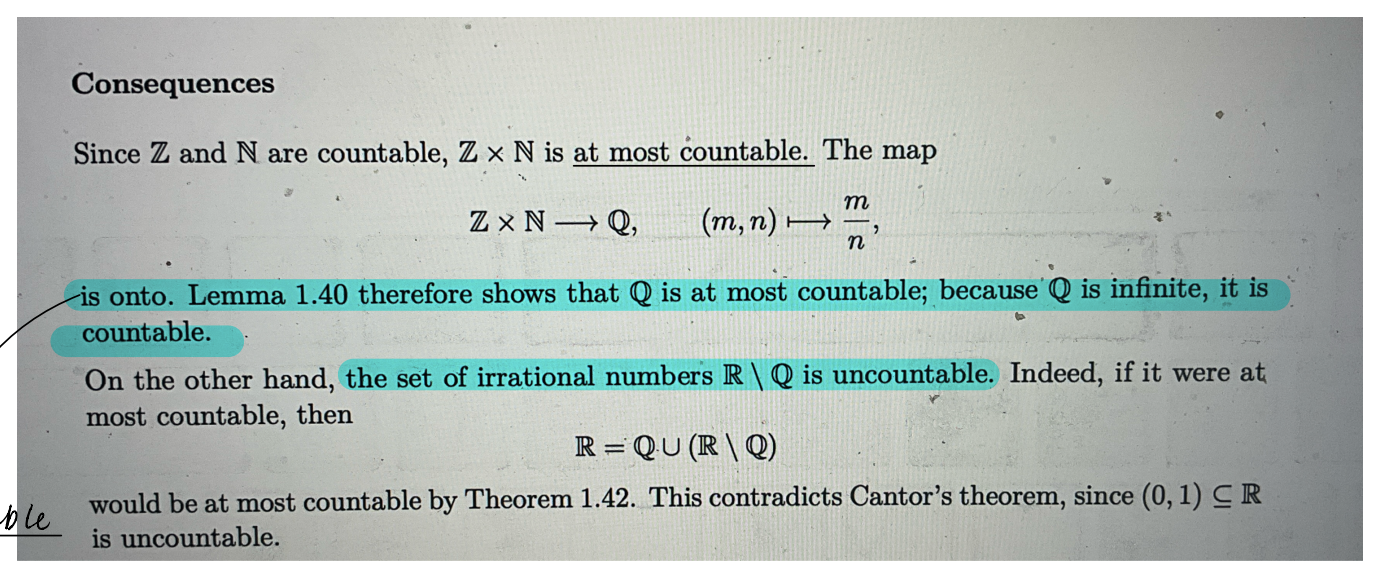
if $E \neq \emptyset$. E is at most countable if and only if there is a surjection $g : \mathbb{N} \to E$.

Let $A \subseteq B$:
(1) if B is at most countable. then A is at most countable
(2) if A is uncountable. then B is uncountable
proof : (1) if $A = \emptyset$. A is at most countable trivially
if $A \neq \emptyset$: there is a surjection $g : \mathbb{N} \to B$
⇓
let $h(n) := \begin{cases} g(n), & g(n) \in A \\ a_0, & g(n) \notin A \end{cases}$
where a_0 is an arbitrary element of A
Hence h is a surjection : $\mathbb{N} \to A$
(2) is (1)'s contrapositive so (1) true ⇒ (2) true

Theorem 1.42 :
Let $A_1, A_2 ..$ be at most countable set
(i) $A_1 \times A_2$ is at most countable
(ii) $\bigcup_{j=1}^{\infty} A_j$ is at most countable
proof :
① introduce a diagonal enumeration
$T_r = r(r+1)/2$
for each $n \in \mathbb{N}$. there is a unique r : $T_{r-1} < n \leq T_r$
Take the bijection : $\mathbb{N} \to \mathbb{N} \times \mathbb{N}$
$n \to (n - T_{r-1}, r + 1 - (n - T_{r-1}))$ ⇒ (1,1), (1,2), (2,1), (1,3), (2,2), (3,1) ..



(i) if A_1 or A_2 is empty. the assertion is true
Otherwise : there exists $f : \mathbb{N} \to A_1$ $g : \mathbb{N} \to A_2$.
take ϕ : $\mathbb{N} \times \mathbb{N} \to A_1 \times A_2$: $\phi(m,n) = (f(m), g(n))$
ϕ is onto. Composing ϕ with a diagonal enumeration $\mathbb{N} \to \mathbb{N} \times \mathbb{N}$
⇒ an onto map : $\mathbb{N} \to A_1 \times A_2$
Hence $A_1 \times A_2$ is at most countable
(ii) write $E = \bigcup_{j \geq 1} A_j$. if $E = \emptyset$. trivially true
otherwise. choose $e_0 \in E$., $\forall$ nonempty A_j, choose a surjection $\mathbb{N} \to A_j$.
$\forall$ empty A_j. set $f_j(n) = e_0$, $\forall n$
Then we can form a map $F : \mathbb{N} \times \mathbb{N} \to E$. $F(j, n) = f_j(n)$
then use the diagonal enumeration on F. ⇒ $\mathbb{N} \to E$ ⇒ at most countable.



Consequences

Since $\mathbb{Z}$ and $\mathbb{N}$ are countable, $\mathbb{Z} \times \mathbb{N}$ is at most countable. The map

$$\mathbb{Z} \times \mathbb{N} \longrightarrow \mathbb{Q}, \qquad (m,n) \longmapsto \frac{m}{n},$$

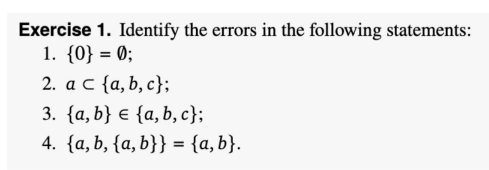
is onto. Lemma 1.40 therefore shows that $\mathbb{Q}$ is at most countable; because $\mathbb{Q}$ is infinite, it is countable.

On the other hand, the set of irrational numbers $\mathbb{R} \setminus \mathbb{Q}$ is uncountable. Indeed, if it were at most countable, then

$$\mathbb{R} = \mathbb{Q} \cup (\mathbb{R} \setminus \mathbb{Q})$$

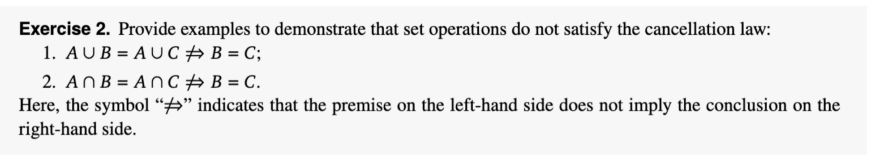
would be at most countable by Theorem 1.42. This contradicts Cantor's theorem, since $(0,1) \subseteq \mathbb{R}$ is uncountable.

$\mathbb{Z} \times \mathbb{N}$ is at most countable
Surjection : $\mathbb{N} \to \mathbb{Z} \times \mathbb{N}$
Surjection : $\mathbb{Z} \times \mathbb{N} \to \mathbb{Q}$ ⇒ $\mathbb{N} \to \mathbb{Q}$ has a surjection



Exercise 1. Identify the errors in the following statements:
1. $\{0\} = \emptyset$;
2. $a \subset \{a, b, c\}$;
3. $\{a, b\} \in \{a, b, c\}$;
4. $\{a, b, \{a, b\}\} = \{a, b\}$.

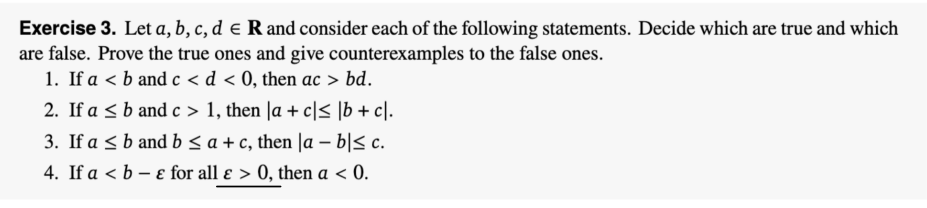
(1) {0} is a non-empty set containing element 0.
(2) ∈ (3) ⊆
(4) $\{a, b\} \subseteq \{a, b, \{a, b\}\}$.



Exercise 2. Provide examples to demonstrate that set operations do not satisfy the cancellation law:
1. $A \cup B = A \cup C \not\Rightarrow B = C$;
2. $A \cap B = A \cap C \not\Rightarrow B = C$.

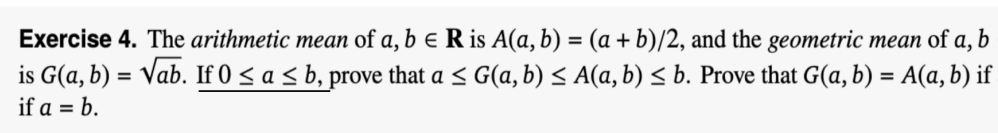
Here, the symbol "$\not\Rightarrow$" indicates that the premise on the left-hand side does not imply the conclusion on the right-hand side.

1. $A = \{a, b, c\}$ $b = \{b\}$. $c = \{c\}$.
2. $A = \{a\}$. $b = \{a, b\}$. $c = \{a, c\}$



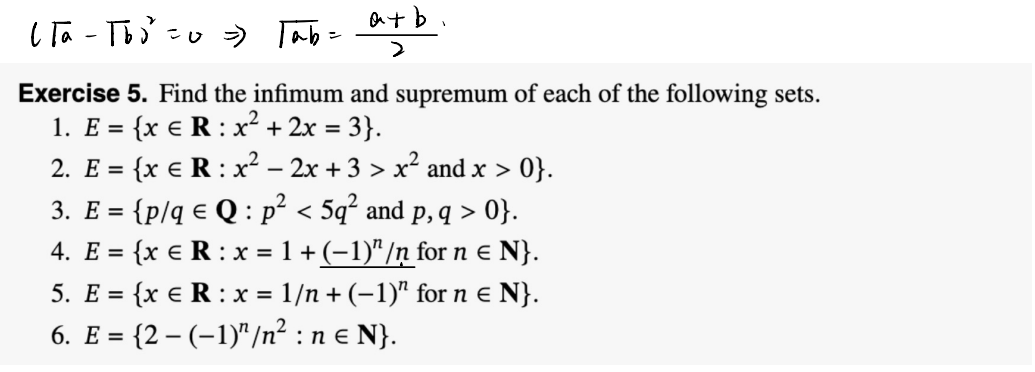
Exercise 3. Let $a, b, c, d \in \mathbf{R}$ and consider each of the following statements. Decide which are true and which are false. Prove the true ones and give counterexamples to the false ones.
1. If $a < b$ and $c < d < 0$, then $ac > bd$.
2. If $a \leq b$ and $c > 1$, then $|a + c| \leq |b + c|$.
3. If $a \leq b$ and $b \leq a + c$, then $|a - b| \leq c$.
4. If $a < b - \varepsilon$ for all $\varepsilon > 0$, then $a < 0$.

(1) false : $a = 1$. $b = 2$. $c = -4$ $d = -1$
$ac = -4 < -2 = bd$
(2) false. $a = -100$ $b = 1$ $c = 2$.
$|a + c| = 98$. $|a + b| = 3$
(3). $a \leq b \leq a + c \Rightarrow 0 \leq b - a \leq c \Rightarrow |a - b| \leq c$ true.
(4). if $a, b \in \mathbb{R}$. then : let $\varepsilon > \max\{0, b - a\}$.
⇒ $\varepsilon > b - a$ and $\varepsilon > 0$. ⇒ Hypothesis cannot hold. ⇒ vacuously true
if $a, b \in \overline{\mathbb{R}} = [-\infty, +\infty]$. let $b = +\infty$. $a = 1$. the $b - \varepsilon = +\infty$ for all ε. $b - \varepsilon > a \not\Rightarrow a < 0$



Exercise 4. The *arithmetic mean* of $a, b \in \mathbf{R}$ is $A(a,b) = (a+b)/2$, and the *geometric mean* of $a, b \in [0, \infty)$ is $G(a,b) = \sqrt{ab}$. If $0 \leq a \leq b$, prove that $a \leq G(a,b) \leq A(a,b) \leq b$. Prove that $G(a,b) = A(a,b)$ if and only if $a = b$.

$G(a,b) = \sqrt{ab} \leq \sqrt{b \cdot b} = b$. $A(a,b) = \frac{a+b}{2} \leq \frac{b+b}{2} = b$.
$\sqrt{ab} \geq \sqrt{a \cdot a} = a$. $= \frac{a+b}{2} \geq \frac{a+a}{2} = a$.
Since : $(\sqrt{a} - \sqrt{b})^2 \geq 0$.
$a + b \geq 2\sqrt{ab} \Rightarrow \sqrt{ab} \leq \frac{a+b}{2} \Rightarrow G(a,b) \leq A(a,b)$.
if $G(a,b) = A(a,b)$. :
$\sqrt{ab} = \frac{a+b}{2} \Rightarrow a - 2\sqrt{ab} + b = 0 \Rightarrow (\sqrt{a} - \sqrt{b})^2 = 0 \Rightarrow \sqrt{a} = \sqrt{b} \Rightarrow a = b$.
if $a = b$.
$(\sqrt{a} - \sqrt{b})^2 = 0 \Rightarrow \sqrt{ab} = \frac{a+b}{2}$.



Exercise 5. Find the infimum and supremum of each of the following sets.
1. $E = \{x \in \mathbf{R} : x^2 + 2x = 3\}$.
2. $E = \{x \in \mathbf{R} : x^2 - 2x + 3 > x^2 \text{ and } x > 0\}$.
3. $E = \{p/q \in \mathbf{Q} : p^2 < 5q^2 \text{ and } p, q > 0\}$.
4. $E = \{x \in \mathbf{R} : x = 1 + (-1)^n/n \text{ for } n \in \mathbf{N}\}$.
5. $E = \{x \in \mathbf{R} : x = 1/n + (-1)^n \text{ for } n \in \mathbf{N}\}$.
6. $E = \{2 - (-1)^n/n^2 : n \in \mathbf{N}\}$.

(1) $E = \{-3, 1\}$. ⇒ $\inf E = -3$. $\sup E = 1$
(2) $x^2 - 2x + 3 > x^2$. ⇒ $2x < 3$ ⇒ $x < \frac{3}{2}$
∴ $E = (0, \frac{3}{2})$. $\inf E = 0$. $\sup E = \frac{3}{2}$
(3) $\frac{p}{q} < \sqrt{5}$ and $\frac{p}{q} > 0$.
⇒ $E = (0, \sqrt{5})$ $\inf E = 0$. $\sup E = \sqrt{5}$
(4). for even n : $x = 1 + \frac{1}{n}$, $n = 2, 4, 6 \cdots$
$x \in (1, \frac{3}{2}]$
for odd n : $x = 1 - \frac{1}{n}$, $n = 1, 3, 5 \cdots$
$x \in [0, 1)$.
∴ $\sup E = \frac{3}{2}$. $\inf E = 0$
(5). for even n : $x = \frac{1}{n} + 1$. $x = 2, 4, 6 \cdots$
$x \in (1, \frac{3}{2}]$.
for odd n : $x = \frac{1}{n} - 1$. $x = 1, 3, 5 \cdots$
$x \in (-1, 0]$.
⇒ $\inf E = -1$. $\sup E = \frac{3}{2}$
(6). for even n : $2 - (-1)^n/n^2 = 2 - \frac{1}{n^2} \in [\frac{7}{4}, 2)$
for odd n : $2 - (-1)^n/n^2 = 2 + \frac{1}{n^2} \in (2, 3]$.
∴ $\inf E = \frac{7}{4}$. $\sup E = 3$.

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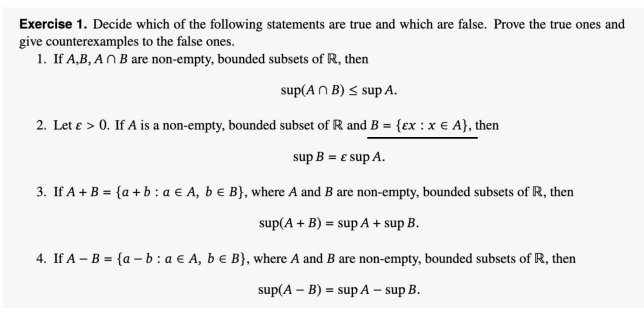
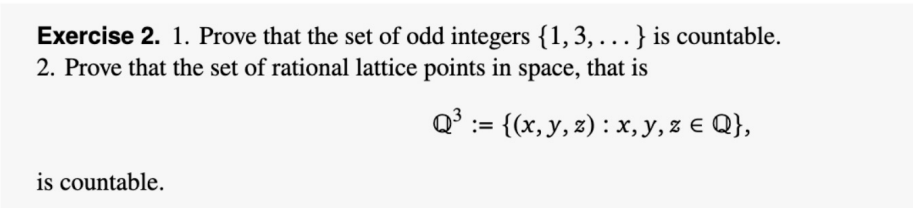


Exercise 1. Decide which of the following statements are true and which are false. Prove the true ones and give counterexamples to the false ones.
1. If $A, B, A \cap B$ are non-empty, bounded subsets of $\mathbb{R}$, then
$$\sup(A \cap B) \leq \sup A.$$
2. Let $\varepsilon > 0$. If A is a non-empty, bounded subset of $\mathbb{R}$ and $B = \{\varepsilon x : x \in A\}$, then
$$\sup B = \varepsilon \sup A.$$
3. If $A + B = \{a + b : a \in A, b \in B\}$, where A and B are non-empty, bounded subsets of $\mathbb{R}$, then
$$\sup(A + B) = \sup A + \sup B.$$
4. If $A - B = \{a - b : a \in A, b \in B\}$, where A and B are non-empty, bounded subsets of $\mathbb{R}$, then
$$\sup(A - B) = \sup A - \sup B.$$

(1) $A = \{-1, 1\}$. $B = \{1\}$.
$\sup(A \cap B) = 1 > -1 = \sup A$
(2). $\forall x \in A$. $x \leq \sup A$. $\varepsilon x \leq \varepsilon \sup A$. ⇒ $\varepsilon \sup A$ is UB of B. ① $\varepsilon \sup A \geq \sup B$.
$\forall x \in A$. $\varepsilon x \in B \Rightarrow \varepsilon x \leq \sup B$, $x \leq \frac{\sup B}{\varepsilon} \Rightarrow \frac{\sup B}{\varepsilon}$ is UB of A $\frac{\sup B}{\varepsilon} \geq \sup A$
⇒ ② $\varepsilon \sup A \leq \sup B$
①, ② : $\sup B = \varepsilon \sup A$
(3). $\forall a \in A$ $b \in B$. $a \leq \sup A$ $b \leq \sup B$.
$\sup A + \sup B \geq \sup(A + B)$

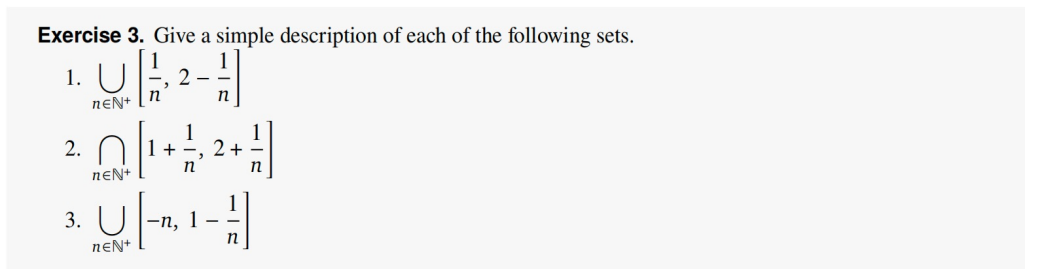
Suppose : $\sup A + \sup B > \sup(A + B)$.
$\sup A > \sup(A + B) - \sup B$,
$\exists a_0 \in A$. $a_0 > \sup(A + B) - \sup B$.
⇒ $\sup B > \sup(A + B) - a_0$.
$\exists b_0 \in B$: $b_0 > \sup(A + B) - a_0$.
$a_0 + b_0 > \sup(A + B)$
☆

Since $a_0 + b_0 \in A + B$. $a_0 + b_0 \leq \sup(A + B)$. a Contradiction.
∴ $\sup A + \sup B = \sup(A + B)$
(4). $A = \{1\}$. $B = \{-1, 1\}$.
$\sup(A - B) = 2 \neq \sup A - \sup B = 0$



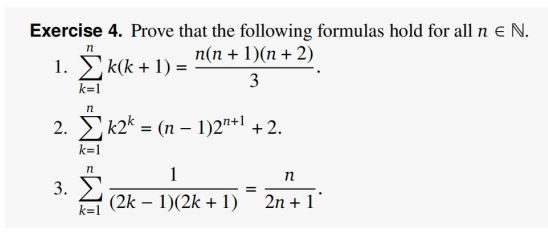
Exercise 2. 1. Prove that the set of odd integers $\{1, 3, \dots\}$ is countable.
2. Prove that the set of rational lattice points in space, that is
$$\mathbb{Q}^3 := \{(x, y, z) : x, y, z \in \mathbb{Q}\},$$
is countable.

(1) let $E = \{1, 3, 5 \cdots\}$. $f : \mathbb{N} \to E$: $f(n) = 2n - 1$. f is a bijection.
(2) Q is at most countable
⇒ $Q \times Q$ is at most countable
⇒ $(Q \times Q) \times Q = Q^3$ is at most countable



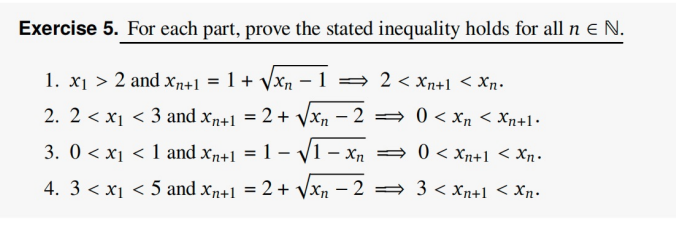
Exercise 3. Give a simple description of each of the following sets.
1. $\bigcup_{n \in \mathbb{N}^+} \left[\frac{1}{n}, 2 - \frac{1}{n}\right]$
2. $\bigcap_{n \in \mathbb{N}^+} \left[1 + \frac{1}{n}, 2 + \frac{1}{n}\right]$
3. $\bigcup_{n \in \mathbb{N}^+} \left[-n, 1 - \frac{1}{n}\right]$

(1) $\lim_{n \to \infty} \frac{1}{n} = 0$. $\lim_{n \to \infty} (2 - \frac{1}{n}) = 2$
$\bigcup_{n \in \mathbb{N}^+} [\frac{1}{n}, 2 - \frac{1}{n}] = [0, 2]$
(2) $\frac{1}{n} \leq 1$ for $n \in \mathbb{N}^+$.
$1 + \frac{1}{n} \leq 2$. $2 \leq 2 + \frac{1}{n} \leq 3$
$\bigcap_{n \in \mathbb{N}^+} [1 + \frac{1}{n}, 2 + \frac{1}{n}] = \{2\}$.
(3). $\bigcup_{n \in \mathbb{N}^+} [-n, 1 - \frac{1}{n}]$
$-n \to -\infty$ as $n \to +\infty$. $1 - \frac{1}{n} < 1$ for all $n \in \mathbb{N}^+$.
∴ $\bigcup_{n \in \mathbb{N}^+} [-n, 1 - \frac{1}{n}] = (-\infty, 1)$



Exercise 4. Prove that the following formulas hold for all $n \in \mathbb{N}$.
1. $\sum_{k=1}^{n} k(k+1) = \frac{n(n+1)(n+2)}{3}$.
2. $\sum_{k=1}^{n} k2^k = (n-1)2^{n+1} + 2$.
3. $\sum_{k=1}^{n} \frac{1}{(2k-1)(2k+1)} = \frac{n}{2n+1}$.

(1). the formula holds for n = 1 : $\sum_{k=1}^{1} k(k+1) = 2 = \frac{1(1+1)(1+2)}{3}$
if the formula holds for n. then :
$\sum_{k=1}^{n+1} k(k+1) = (n+1)(n+2) + \sum_{k=1}^{n} k(k+1) = (n+1)(n+2) + \frac{n(n+1)(n+2)}{3}$
⇒ the formula holds for n+1
$= \frac{(n+3)(n+1)(n+2)}{3}$
$= \frac{(n+1)[(n+1)+1][(n+2)+1]}{3}$
(2) the formula holds for n = 1 :
if the formula holds for n. then :
$\sum_{k=1}^{n+1} k2^k = (n+1)2^{n+1} + (n-1)2^{n+1} + 2$.
$= 2n \cdot 2^{n+1} + 2$.
$= n \cdot 2^{n+2} + 2$ ⇒ holds for n+1
(3). the formula holds for n = 1 :
if the formula holds for n. then :
$\sum_{k=1}^{n+1} \frac{1}{(2k-1)(2k+1)} = \frac{1}{(2n+1)(2n+3)} + \frac{n}{2n+1} = \frac{1 + n(2n+3)}{(2n+1)(2n+3)} = \frac{2n^2 + 3n + 1}{(2n+1)(2n+3)} = \frac{n+1}{2(n+1)+1}$ ⇒ holds for n+1



Exercise 5. For each part, prove the stated inequality holds for all $n \in \mathbb{N}$.
1. $x_1 > 2$ and $x_{n+1} = 1 + \sqrt{x_n - 1} \implies 2 < x_{n+1} < x_n$.
2. $2 < x_1 < 3$ and $x_{n+1} = 2 + \sqrt{x_n - 2} \implies 0 < x_n < x_{n+1}$.
3. $0 < x_1 < 1$ and $x_{n+1} = 1 - \sqrt{1 - x_n} \implies 0 < x_{n+1} < x_n$.
4. $3 < x_1 < 5$ and $x_{n+1} = 2 + \sqrt{x_n - 2} \implies 3 < x_{n+1} < x_n$.

Tips: We could firstly prove the inequalities as follows:
1. If $a > 2$ and $b = 1 + \sqrt{a - 1}$, then $2 < b < a$.
2. If $2 < a < 3$ and $b = 2 + \sqrt{a - 2}$, then $0 < a < b$.
3. If $0 < a < 1$ and $b = 1 - \sqrt{1 - a}$, then $0 < b < a$.
4. If $3 < a < 5$ and $b = 2 + \sqrt{a - 2}$, then $3 < b < a$.

$b = 1 + \sqrt{a-1} > 1 + \sqrt{2-1} = 2$., $b = 1 + \sqrt{a-1} < 1 + (a-1) = a$
$a - 2 < \sqrt{a-2} \Rightarrow a < \sqrt{a-2} + 2 = b$.
$1 - a < \sqrt{1-a} \Rightarrow a > 1 - \sqrt{1-a} = b$.
$a - 2 > \sqrt{a-2} \Rightarrow a > \sqrt{a-2} + 2 = b$

then : proof 1−4's eq holds for n = 1, then by induction. eq holds for all n.